Ricardo (Zhicong) Xie

Simulation / Analysis Intern | FEA, CFD and Validation

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Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student specialising in structural FEA and thermal-fluid CFD, with project work in Abaqus/Standard, ANSYS Fluent/CFX and model credibility checks.
- I support engineering teams by delivering analysis with clear load cases, boundary conditions, mesh checks, mass/heat balance review, result interpretation and documented model limitations.
- Project screenshots and source links: https://zxie588-alt.github.io

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Finite Element Analysis, Computational Fluid Dynamics, Composite Materials, Thermal Modelling

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles, compared drawing intent with cut parts, and worked around CNC workflow, workholding, chip formation and shop-floor safety awareness.
- Engineering practice: interpreted force-displacement test output, practised robotic arm control, and acted as a small-group lead explaining automotive seat/interior structures to peers.
- Technical visual communication: former student news-department lead with photography/event documentation experience; professional working English and native Mandarin.

PROJECT WORK

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Developed a shell FEA model of a composite Formula SAE front wing in Abaqus/Standard using S4R elements, layered laminate definitions and local mesh refinement at high-stress attachment points.
- Converted aero downforce/drag, lateral cornering and 4.1 G bump conditions into structural load cases for strength and stiffness checks.
- Evaluated ply-level stress utilisation against fatigue and ultimate failure criteria, achieving about 74% peak utilisation under the governing 4.1 G bump case with SF = 1.5.

Thermal-Fluid Simulation & Validation - Solar Chimney Case Study

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD validation workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Validated model credibility through mesh-sensitivity checks, global mass/heat balance review, Reynolds/Rayleigh scaling and solver-behaviour comparison before trusting velocity results.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, validation workflow

- Designed a SolidWorks enclosure design package with screw bosses, gasket stack-up, PCB and battery envelopes, cable/strap mounts and service access.
- Applied DFM principles to improve enclosure features for CNC machining review and 3D-print prototype preparation.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built prototype firmware in Keil C with sensor-driver abstraction, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Delivered a small residential MVHR/ERV design package for a New Zealand building-services workflow, covering layout intent, BIM/CAD coordination, airflow scheduling and review notes.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Built a Revit 2026 MEP coordination model with a one-level residential layout, MVHR unit, supply/extract routes, diffusers/grilles, airflow schedule, three portfolio sheets, five drawing-review comments and a simple clash-check note; published RVT and screenshots through the portfolio.
- Sized the system for 60 L/s balanced supply/extract in normal operation and 85 L/s boost extract mode, calculating external static pressure targets of about 80 Pa and 120 Pa.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; developed a commissioning test plan covering CO2, humidity, airflow and fan-power measurements.

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering workstream for a 5-person team study covering a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Delivered a feasibility-report structure with trade-off matrices, risk logic and verification-gate framework across orbit/coverage, power, thermal rejection, communications, launch mass, cost and sustainability.

TECHNICAL SKILLS

- Proficient in: Abaqus/Standard, ANSYS Fluent/CFX, shell FEA, CFD, conjugate heat transfer and natural-convection modelling
- Proficient in: load-case definition, boundary conditions, mesh-sensitivity checks, assumptions log, result interpretation and model-limit documentation
- Familiar with: composite failure criteria, ply utilisation checks, Rayleigh/Reynolds scaling, global mass/heat balance validation